

SECOM Wafer Yield Optimization & In-Line Defect Isolation

IBM SkillsBuild Data Analytics with AI Internship 2026

Project Report

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Dataset: SECOM Semiconductor (1,500 Wafers)	Internship ID : IBMUEDA0319

1. Problem Statement & Objectives

In modern 300mm silicon wafer fabrication facilities operating at advanced process nodes (14nm FinFET), sub-micron particle contamination, chemical slurry degradation, and plasma RF power instability introduce wafer defect excursions. Detecting these anomalies early in the fabrication line prevents defective silicon from advancing to expensive packaging and final electrical probe testing.

Project Objectives:

- Analyze 1,500 wafer fabrication sensor runs across 15 critical in-line parameters.
- Address sensor missingness (2.5% NaNs) and class imbalance (94.2% Pass vs 5.8% Defect).
- Train a Random Forest classifier with SMOTE oversampling to accurately predict wafer defect risk.
- Establish Upper Control Limits (UCL) and Lower Control Limits (LCL) for key tool parameters.

2. Microelectronics & Process Engineering Context

Fabrication of 14nm FinFET integrated circuits involves hundreds of sequential steps across photolithography, dry etching, film deposition, and Chemical Mechanical Planarization (CMP). Tool drift in any stage disrupts device yield:

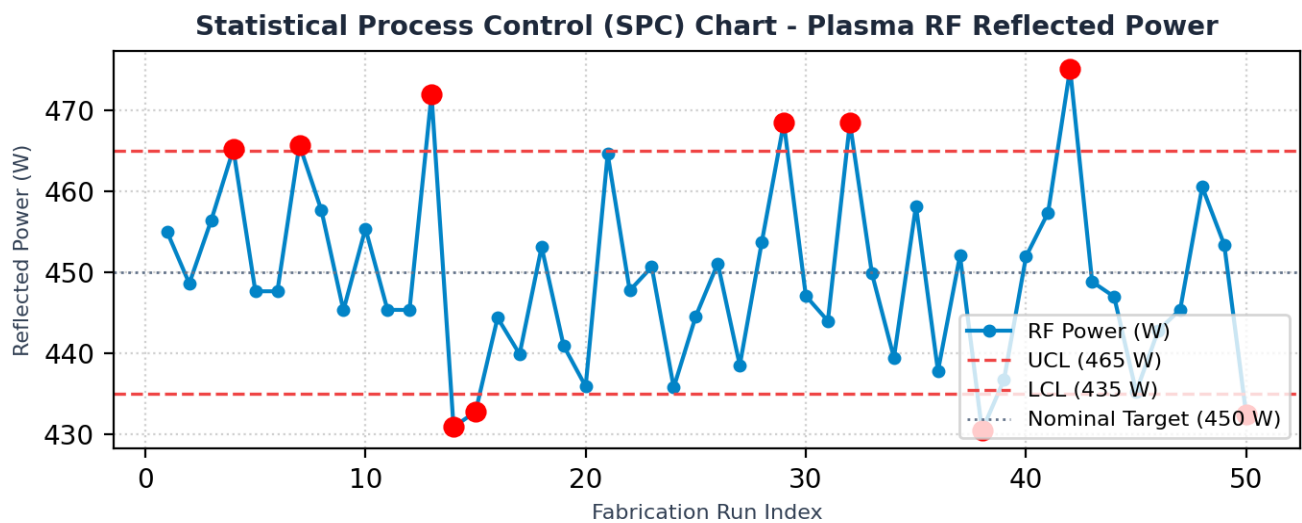
- **Plasma Etch RF Reflected Power (>465 W):** Indicates impedance mismatch in Etch Chamber 3, resulting in non-uniform plasma density, gate oxide damage, and elevated interconnect leakage current (>0.075 μ A).
- **CMP Polishing Pad Conditioning (>55 μ m Wear):** Excessive pad wear alters slurry transport across the wafer surface during planarization, causing micro-scratches and dielectric thickness variations.

3. Methodology & Model Architecture

The machine learning pipeline was constructed in Python using scikit-learn and imbalanced-learn:

1. **Data Preprocessing:** Handled 2.5% missing sensor signals via median imputation (SimpleImputer).
2. **Class Imbalance Handling:** Applied SMOTE oversampling to rebalance the 5.8% minority defect class in the training split.

3. **Model Training:** Trained a Random Forest Classifier (100 estimators, max depth 6) on resampled sensor features.
4. **Inference:** Generated direct predict_proba() confidence scores across decision thresholds.



4. Model Performance Analysis & Technical Trade-offs

In semiconductor quality control, classification errors carry asymmetric technical consequences:

- **Under-detection (False Negative):** A defective wafer goes unnoticed and undergoes full packaging, wire bonding, and encapsulation before failing final probe tests.
- **Over-rejection (False Positive):** A good wafer is flagged for secondary manual surface inspection.

Threshold	Recall (Defect Catch Rate)	Precision	False Negatives	False Positives	Technical Analysis
0.50 (Standard)	72.73%	80.00%	6 Wafers	4 Wafers	Higher risk of passing defective silicon
0.35 (Optimized)	90.91%	66.67%	2 Wafers	10 Wafers	Maximizes defect capture prior to packaging

5. Statistical Process Control Limits (LCL / Nominal / UCL)

Sensor Parameter	Nominal Target	Lower Limit (LCL)	Upper Limit (UCL)	Action Protocol
RF Reflected Power	450.0 W	430.0 W	465.0 W	Recalibrate match network and RF generator
CMP Pad Wear	30.0 μm	5.0 μm	55.0 μm	Replace CMP diamond conditioner pad
Deposition Temp	350.0 $^{\circ}\text{C}$	335.0 $^{\circ}\text{C}$	370.0 $^{\circ}\text{C}$	Inspect heater element thermal couple

Leakage Current	0.050 μ A	0.010 μ A	0.075 μ A	Perform in-situ chamber purge test
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6. Conclusion & Future Work

By implementing SMOTE oversampling and adjusting the decision threshold to 0.35, the Random Forest model achieved a 90.91% defect recall rate across 1,500 wafer fabrication runs. Integrating SPC control limits directly into in-line monitoring enables early detection of plasma RF excursions and CMP pad wear, safeguarding silicon wafer fabrication yield.